PW EARNERS

DSA in C++

Daily Practice Problems (DPP) — 4

Topic: Pattern Printing

For Q1 to Q10: Analyze the nested loop architectures and predict the exact output ***on your own; do not predict the output pattern by running the code****. If a question contains multiple choice options, select the correct one.*

Q1.

```cpp
#include<iostream>
using namespace std;
int main(){
    for(int i = 1; i <= 3; i++) {
        for(int j = 1; j <= 2; j++) {
            cout << i << j << " ";
        }
    }
}
```

Output:

. . . .

Q2.

```cpp
#include<iostream>
using namespace std;
int main(){
    for(int i = 1; i <= 4; i++) {
        for(int j = 4; j >= i; j--) {
            cout << "*";
        }
        cout << endl;
    }
}
```

Which pattern is generated by this block?

A. A right-angled triangle pointing downwards (4 stars on top row, decreasing down to 1).

B. A right-angled triangle pointing upwards (1 star on top row, increasing down to 4).

C. A square grid containing a total of 16 stars.

D. A vertically inverted mirror pyramid structure.

Correct Option: . . . .

Q3.

```cpp
#include<iostream>
using namespace std;
int main(){
    int n = 3;
    for(int i = 1; i <= n; i++) {
        char ch = 'A';
        for(int j = 1; j <= i; j++) {
            cout << ch++;
        }
        cout << endl;
    }
}
```

Output:

....

Q4.

```cpp
#include<iostream>
using namespace std;
int main(){
    for(int i = 1; i <= 3; i++) {
        for(int j = 1; j <= i; j++) {
            if((i + j) % 2 == 0) cout << "1";
            else cout << "0";
        }
        cout << endl;
    }
}
```

Output:

....

Q5.

```cpp
#include<iostream>
using namespace std;
int main(){
    int count = 1;
    for(int i = 1; i <= 3; i++) {
        for(int j = 1; j <= 3; j++) {
            if(i == j) {
                cout << "X";
            } else {
                cout << count++;
            }
        }
        cout << endl;
    }
}
```

Output:

....

Q6.

```cpp
#include<iostream>
using namespace std;
int main(){
    for(int i = 1; i <= 3; i++) {
        int val = i;
        for(int j = 1; j <= 3; j++) {
            cout << val << " ";
            val += 2;
        }
        cout << endl;
    }
}
```

Output:

....

Q7.

```cpp
#include<iostream>
using namespace std;
int main(){
    for(int i = 0; i < 4; i++) {
        for(int j = 0; j < 4; j++) {
            if(i == 0 || i == 3 || j == 0 || j == 3)
                cout << "#";
            else
                cout << " ";
        }
        cout << endl;
    }
}
```

What configuration does this coordinate filtering display?

A. A filled square consisting entirely of hash symbols.

B. A cross pattern matching the shape of a large central addition plus symbol.

C. A hollow perimeter border shell framework enclosing a blank inner space.

D. Two crossing diagonal line structures meeting perfectly in the middle coordinates.

Correct Option: ....

Q8.

```cpp
#include<iostream>
using namespace std;
int main(){
    int n = 3;
    for(int i = 1; i <= n; i++) {
        for(int space = 1; space <= n - i; space++) cout << " ";
        for(int j = 1; j <= i; j++) cout << "*";
        cout << endl;
```

```
    }
}
```

Output:

. . . .

Q9. ***(Crazy Question 1)***

```
#include<iostream>
using namespace std;
int main(){
    int n = 4;
    for(int i = 1; i <= n; i++) {
        int x = 1;
        for(int j = 1; j <= n; j++) {
            if(j <= n - i) {
                cout << " ";
            } else {
                cout << x;
                x = !x;
            }
        }
        cout << endl;
    }
}
```

Output:

. . . .

Q10. ***(Crazy Question 2)***

```
#include<iostream>
using namespace std;
int main(){
    int k = 1;
    for(int i = 1; i <= 3; i++) {
        for(int j = 1; j <= i; i++) {
            cout << k++ << " ";
            if(k > 4) break;
        }
        if(k > 4) break;
        cout << endl;
    }
}
```

Output:

. . . .

*For Q11 to Q20: Write clean, modular, and compilable C++ programs. Use proper nested loop structures. Assume user inputs an integer **n** denoting structural constraints unless specified otherwise.*

Q11. Take an odd integer `n` as input from the user ($n \geq 3$). Write a program to print a **Plus**

inside Square Boundary pattern.
Sample — Input: `n = 15` ⇒ Output:

```
***************
*      *      *
*      *      *
*      *      *
*      *      *
*      *      *
*      *      *
***************
*      *      *
*      *      *
*      *      *
*      *      *
*      *      *
*      *      *
***************
```

Q12. Take a positive integer `n` as input. Write a program to print a **Repeated Number Inverted Right Triangle**. Each row should display the same number, corresponding to its current descending row count.
Sample — Input: `n = 4` ⇒ Output:

```
4 4 4 4
3 3 3
2 2
1
```

Q13. Write a program to print an **Alternating Row Offset Pattern** for a given integer `n` as shown below.
Sample — Input: `n = 5` ⇒ Output:

```
1 1 1 1 1
 2 2 2 2
3 3 3 3 3
 4 4 4 4
5 5 5 5 5
```

Q14. Write a program that prints an **Inscribed Hollow Diamond** pattern. The pattern consists of a solid rectangle of stars with a hollow diamond shape in the center.
Sample — Input: `n = 5` ⇒ Output:

```
**********
****  ****
***    ***
**      **
*        *
*        *
**      **
***    ***
****  ****
**********
```

Q15. Write a program to print a symmetric **Cross Alphabet Pattern** for a given integer `n`.
Sample — Input: `n = 4` ⇒ Output:

```
A       A
 B     B
  C   C
   D D
  C   C
 B     B
A       A
```

Q16. Create a program that takes an odd integer `n` and prints a **Square with Diagonals**. The grid should display a solid outer boundary along with intersecting major and minor diagonals.
Sample — Input: `n = 8` ⇒ Output:

```
********
**    **
* *  * *
*  **  *
*  **  *
* *  * *
**    **
********
```

Q17. Write a program to construct an **Alphabet Palindrome Pyramid**. For a given `n`, each row should print letters sequentially increasing to a peak character, and then symmetrically decreasing back to 'A'. Use precise spacing to center the pyramid.
Sample — Input: `n = 4` ⇒ Output:

```
   A
  ABA
 ABCBA
ABCDCBA
```

Q18. Design a program to generate a geometric **Hollow Star Diamond**. (Assume `n` defines the number of rows in the top half including the center).
Sample — Input: `n = 3` ⇒ Output:

```
  *
 * *
*   *
 * *
  *
```

Q19. Write a program to print the classic **Butterfly Pattern**. This layout consists of two mirrored growing right triangles on the top half, and two mirrored shrinking right triangles on the bottom half, separated by precise hollow space.
Sample — Input: `n = 4` (Half rows) ⇒ Output:

```
*      *
**    **
***  ***
********
********
***  ***
**    **
*      *
```

Q20. Write a program to print an **Expanding Number Diamond** pattern for a given integer `n`.
Sample — Input: `n = 4` ⇒ Output:

```
      1
    2 3 2
  3 4 5 4 3
4 5 6 7 6 5 4
  3 4 5 4 3
    2 3 2
      1
```

All the best! Keep Coding ♡